

Create a registration function for the Mars images

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The first step when stitching images together is to register pairs of images. Here, you will test your registration function. If your function does not pass the assessment the Registration Estimator app and refine your approach.

1. Use the Registration Estimator app to create a registration function for the provided images from the Mars Curiosity Rover.
2. Copy the contents of the generated function into the code box below. **Do not** copy the function signature line. Do make sure you include the final **end** keyword
3. Press submit to test if your function is producing the expected result.

The grader is checking that your registration function can produce a result close to the expected solution registration function. You will still need to correctly warp and crop images together.

Do not change the default output field names used by the Registration Estimator app. You can add more output fields if you want for debugging, but the default fields must be present to pass.

Function ?

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function [MOVINGREG] = registerMarsImages(MOVING, FIXED)
2
3 % Default spatial referencing objects
4 fixedRefObj = imref2d(size(FIXED));
5 movingRefObj = imref2d(size(MOVING));
6
7 % Detect SURF features
8 fixedPoints = detectSURFFeatures(FIXED, 'MetricThreshold', 900.000000, 'NumOctaves', 2, 'NumScaleLevels', 4);
9 movingPoints = detectSURFFeatures(MOVING, 'MetricThreshold', 900.000000, 'NumOctaves', 2, 'NumScaleLevels', 4);
10
11 % Extract features
12 [fixedFeatures, fixedValidPoints] = extractFeatures(FIXED, fixedPoints, 'Upright', false);
13 [movingFeatures, movingValidPoints] = extractFeatures(MOVING, movingPoints, 'Upright', false);
14
15 % Match features
16 indexPairs = matchFeatures(fixedFeatures, movingFeatures, 'MatchThreshold', 50.000000, 'MaxRatio', 0.500000);
17 fixedMatchedPoints = fixedValidPoints(indexPairs(:, 1));
18 movingMatchedPoints = movingValidPoints(indexPairs(:, 2));
19 MOVINGREG.FixedMatchedFeatures = fixedMatchedPoints;
20 MOVINGREG.MovingMatchedFeatures = movingMatchedPoints;
21
22 % Apply transformation - Results may not be identical between runs because of the randomized nature of the algorithm
23 tform = estimateGeometricTransform2D(movingMatchedPoints, fixedMatchedPoints, 'projective');
24 MOVINGREG.Transformation = tform;
25 MOVINGREG.RegisteredImage = imwarp(MOVING, movingRefObj, tform, 'OutputView', fixedRefObj, 'SmoothEdges', true);
26
27 % Store spatial referencing object
28 MOVINGREG.SpatialRefObj = fixedRefObj;
29
30 end
```

Code to call your function ?

Reset

```
1 % Load the images
2 fixedImg = imread("sol_03333_opgs_edr_ncam_NLB_693387385EDR_F0921230NCAM00259M_.JPG");
3 leftImg = imread("sol_03333_opgs_edr_ncam_NLB_693387301EDR_F0921230NCAM00259M_.JPG");
4
5 % Register the two images
6 reg = registerMarsImages(leftImg, fixedImg)
7
8 figure; imshow(reg.RegisteredImage)
9
```

Run Function ?

Assessment: All Tests Passed

Submit ?

Does the output structure have the expected fields?

✔ Is the output transformation the correct type?

✔ Do the registered images appear correct?